

Then, (11.68), which is nondifferentiable due to the hinge loss function, can be reformulated in terms of solving the 1-norm soft-margin optimization problem (11.34)–(11.35).

11.4 Multiclass Support Vector Machines

Often, data are derived from more than two classes. In the multiclass situation, $\mathbf{X} \in \mathbb{R}^r$ is a random r -vector chosen for classification purposes and $Y \in \{1, 2, \dots, K\}$ is a class label, where K is the number of classes. Because SVM classifiers are formulated for only two classes, we need to know if (and how) the SVM methodology can be extended to distinguish between $K > 2$ classes. There have been several attempts to define such a multiclass SVM strategy.

11.4.1 Multiclass SVM as a Series of Binary Problems

The standard SVM strategy for a multiclass classification problem (over K classes) has been to reduce it to a series of binary problems. There are different approaches to this strategy:

One-versus-rest: Divide the K -class problem into K binary classification subproblems of the type “ k th class” vs. “not k th class,” $k = 1, 2, \dots, K$. Corresponding to the k th subproblem, a classifier $\hat{f}_k$ is constructed in which the k th class is coded as positive and the union of the other classes is coded as negative. A new $\mathbf{x}$ is then assigned to the class with the largest value of $\hat{f}_k(\mathbf{x})$, $k = 1, 2, \dots, K$, where $\hat{f}_k(\mathbf{x})$ is the optimal SVM solution for the binary problem of the k th class versus the rest.

One-versus-one: Divide the K -class problem into $\binom{K}{2}$ comparisons of all pairs of classes. A classifier $\hat{f}_{jk}$ is constructed by coding the j th class as positive and the k th class as negative, $j, k = 1, 2, \dots, K$, $j \neq k$. Then, for a new $\mathbf{x}$, aggregate the votes for each class and assign $\mathbf{x}$ to the class having the most votes.

Even though these strategies are widely used in practice to resolve multiclass SVM classification problems, one has to be cautious about their use.

In Table 11.3, we report the CV/10 misclassification rates for one-versus-one multiclass SVM applied to the same data sets from Section 8.7. Also listed in Table 11.3 are the values of (C, γ) that yield the minimum misclassification rate for each data set. It is instructive to compare these rates with those in Table 8.7, where we used LDA and QDA. We see that for

TABLE 11.3. Summary of support vector machine (SVM) “one-versus-one” classification results for data sets with more than two classes. Listed are the sample size (n), number of variables (r), and number of classes (K). Also listed for each data set is the 10-fold cross-validation (CV/10) misclassification rates corresponding to the best choice of (C, γ) . The data sets are listed in increasing order of LDA misclassification rates (Table 8.7).

Data Set	n	r	K	SVM-CV/10	C	γ
Wine	178	13	3	0.0169	10^6	8×10^{-8}
Iris	150	4	3	0.0200	100	0.002
Primate scapulae	105	7	5	0.0286	100	0.0002
Shuttle	43,500	8	7	0.0019	10	0.0001
Diabetes	145	5	3	0.0414	100	0.000009
Pendigits	10,992	16	10	0.0031	10	0.0001
E-coli	336	7	8	0.1280	10	1.0
Vehicle	846	18	4	0.1501	600	0.00005
Letter recognition	20,000	16	26	0.0183	50	0.04
Glass	214	9	6	0.0093	10	0.001
Yeast	1,484	8	10	0.3935	10	7.0

the shuttle, diabetes, pendigits, vehicle, letter recognition, glass, and yeast data sets, the SVM method performs better than does the LDA method; for the iris, primate scapulae, and e-coli data sets, the SVM and LDA methods perform about the same; and LDA performs better than does SVM for the wine data set. Thus, neither one-versus-one SVM nor LDA performs uniformly best for all of these data sets.

The one-versus-rest approach is popular for carrying out text categorization tasks, where each document may belong to more than one class. Although it enjoys the optimality property of the SVM method for each binary subproblem, it can yield a different classifier than the Bayes optimal classifier for the multiclass case. Furthermore, the classification success of the one-versus-rest approach depends upon the extent of the class-size imbalance of each subproblem and whether one class dominates all other classes when determining the most-probable class for each new $\mathbf{x}$.

The one-versus-one approach, which uses only those observations belonging to the classes involved in each pairwise comparison, suffers from the problem of having to use smaller samples to train each classifier, which may, in turn, increase the variance of the solution.

11.4.2 A True Multiclass SVM

To construct a true multiclass SVM classifier, we need to consider all K classes, $\Pi_1, \Pi_2, \dots, \Pi_K$, simultaneously, and the classifier has to reduce to

the binary SVM classifier if $K = 2$. Here we describe the construction due to Lee, Lin, and Wahba (2004).

Let $\mathbf{v}_1, \dots, \mathbf{v}_K$ be a sequence of K -vectors, where $\mathbf{v}_k$ has a 1 in the k th position and whose elements sum to zero, $k = 1, 2, \dots, K$; that is, let

$$\begin{aligned}\mathbf{v}_1 &= \left(1, -\frac{1}{K-1}, \dots, -\frac{1}{K-1}\right)^\tau \\ \mathbf{v}_2 &= \left(-\frac{1}{K-1}, 1, \dots, -\frac{1}{K-1}\right)^\tau \\ &\vdots \\ \mathbf{v}_K &= \left(-\frac{1}{K-1}, -\frac{1}{K-1}, \dots, 1\right)^\tau.\end{aligned}$$

Note that if $K = 2$, then $\mathbf{v}_1 = (1, -1)^\tau$ and $\mathbf{v}_2 = (-1, 1)^\tau$. Every $\mathbf{x}_i$ can be labeled as one of these K vectors; that is, $\mathbf{x}_i$ has label $\mathbf{y}_i = \mathbf{v}_k$ if $\mathbf{x}_i \in \Pi_k$, $i = 1, 2, \dots, n$, $k = 1, 2, \dots, K$.

Next, we generalize the separating function $f(\mathbf{x})$ to a K -vector of separating functions,

$$\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_K(\mathbf{x}))^\tau, \quad (11.74)$$

where

$$f_k(\mathbf{x}) = \beta_{0k} + h_k(\mathbf{x}), \quad h_k \in \mathcal{H}_K, \quad k = 1, 2, \dots, K. \quad (11.75)$$

In (11.75), $\mathcal{H}_K$ is a reproducing-kernel Hilbert space (rkhs) spanned by the $\{K(\mathbf{x}_i, \cdot), i = 1, 2, \dots, n\}$. For example, in the linear case, $h_k(\mathbf{x}) = \mathbf{x}^\tau \boldsymbol{\beta}_k$, for some vector of coefficients $\boldsymbol{\beta}_k$. We also assume, for uniqueness, that

$$\sum_{k=1}^K f_k(\mathbf{x}) = 0. \quad (11.76)$$

Let $\mathbf{L}(\mathbf{y}_i)$ be a K -vector with 0 in the k th position if $\mathbf{x}_i \in \Pi_k$, and 1 in all other positions; this vector represents the equal costs of misclassifying $\mathbf{x}_i$ (and allows for an unequal misclassification cost structure if appropriate). If $K = 2$ and $\mathbf{x}_i \in \Pi_1$, then $\mathbf{L}(\mathbf{y}_i) = (0, 1)^\tau$, while if $\mathbf{x}_i \in \Pi_2$, then $\mathbf{L}(\mathbf{y}_i) = (1, 0)^\tau$.

The multiclass generalization of the optimization problem (11.68) is, therefore, to find functions $\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_K(\mathbf{x}))^\tau$ satisfying (11.76) which

$$\text{minimize } I_\lambda(\mathbf{f}, \mathcal{Y}) = \frac{1}{n} \sum_{i=1}^n [\mathbf{L}(\mathbf{y}_i)]^\tau (\mathbf{f}(\mathbf{x}_i) - \mathbf{y}_i)_+ + \frac{\lambda}{2} \sum_{k=1}^K \|h_k\|^2, \quad (11.77)$$

where $(\mathbf{f}(\mathbf{x}_i) - \mathbf{y}_i)_+ = ((f_1(\mathbf{x}_i) - y_{i1})_+, \dots, (f_K(\mathbf{x}_i) - y_{iK})_+)^\tau$ and $\mathcal{Y} = (\mathbf{y}_1, \dots, \mathbf{y}_n)$ is a $(K \times n)$ -matrix.

By setting $K = 2$, we can see that (11.77) is a generalization of (11.68). If $\mathbf{x}_i \in \Pi_1$, then $\mathbf{y}_i = \mathbf{v}_1 = (1, -1)^\tau$, and

$$\begin{aligned} [\mathbf{L}(\mathbf{y}_i)]^\tau (\mathbf{f}(\mathbf{x}_i) - \mathbf{y}_i)_+ &= (0, 1)((f_1(\mathbf{x}_i) - 1)_+, (f_2(\mathbf{x}_i) + 1)_+)^T \\ &= (f_2(\mathbf{x}_i) + 1)_+ \\ &= (1 - f_1(\mathbf{x}_i))_+, \end{aligned} \quad (11.78)$$

while if $\mathbf{x}_i \in \Pi_2$, then $\mathbf{y}_i = \mathbf{v}_2 = (-1, 1)$, and

$$[\mathbf{L}(\mathbf{y}_i)]^\tau (\mathbf{f}(\mathbf{x}_i) - \mathbf{y}_i)_+ = (f_1(\mathbf{x}_i) + 1)_+. \quad (11.79)$$

So, the first term (with f) in (11.68) is identical to the first term (with f_1) in (11.77) when $K = 2$. If we set $K = 2$ in the second term of (11.77), we have that

$$\sum_{k=1}^2 \|h_k\|^2 = \|h_1\|^2 + \|-h_1\|^2 = 2 \|h_1\|^2, \quad (11.80)$$

so that the second terms of (11.68) and (11.77) are identical.

The function $h_k \in \mathcal{H}_K$ can be decomposed into two parts:

$$h_k(\cdot) = \sum_{\ell=1}^n \beta_{\ell k} K(\mathbf{x}_\ell, \cdot) + h_k^\perp(\cdot), \quad (11.81)$$

where the $\{\beta_{\ell k}\}$ are constants and $h_k^\perp(\cdot)$ is an element in the rkhs orthogonal to $\mathcal{H}_K$. Substituting (11.76) into (11.77), then using (11.81), and rearranging terms, we have that

$$f_K(\cdot) = - \sum_{k=1}^{K-1} \beta_{0k} - \sum_{k=1}^{K-1} \sum_{i=1}^n \beta_{ik} K(\mathbf{x}_i, \cdot) - \sum_{k=1}^{K-1} h_k^\perp(\cdot). \quad (11.82)$$

Because $K(\cdot, \cdot)$ is a reproducing kernel,

$$\langle h_k, K(\mathbf{x}_i, \cdot) \rangle = h_k(\mathbf{x}_i), \quad i = 1, 2, \dots, n, \quad (11.83)$$

and so,

$$\begin{aligned} f_k(\mathbf{x}_i) &= \beta_{0k} + h_k(\mathbf{x}_i) \\ &= \beta_{0k} + \langle h_k, K(\mathbf{x}_i, \cdot) \rangle \\ &= \beta_{0k} + \left\langle \sum_{\ell=1}^n \beta_{\ell k} K(\mathbf{x}_\ell, \cdot) + h_k^\perp(\cdot), K(\mathbf{x}_i, \cdot) \right\rangle \\ &= \beta_{0k} + \sum_{\ell=1}^n \beta_{\ell k} K(\mathbf{x}_\ell, \mathbf{x}_i). \end{aligned} \quad (11.84)$$

Note that, for $k = 1, 2, \dots, K-1$,

$$\begin{aligned} \|h_k(\cdot)\|^2 &= \left\| \sum_{\ell=1}^n \beta_{\ell k} K(\mathbf{x}_\ell, \cdot) + h_k^\perp(\cdot) \right\|^2 \\ &= \sum_{\ell=1}^n \sum_{i=1}^n \beta_{\ell k} \beta_{i k} K(\mathbf{x}_\ell, \mathbf{x}_i) + \|h_k^\perp(\cdot)\|^2, \end{aligned} \quad (11.85)$$

and, for $k = K$,

$$\|h_K(\cdot)\|^2 = \left\| \sum_{k=1}^{K-1} \sum_{i=1}^n \beta_{i k} K(\mathbf{x}_i, \cdot) \right\|^2 + \left\| \sum_{k=1}^{K-1} h_k^\perp(\cdot) \right\|^2. \quad (11.86)$$

Thus, to minimize (11.86), we set $h_k^\perp(\cdot) = 0$ for all k .

From (11.84), the zero-sum constraint (11.76) becomes

$$\bar{\beta}_0 + \sum_{\ell=1}^n \bar{\beta}_\ell K(\mathbf{x}_\ell, \cdot) = 0, \quad (11.87)$$

where $\bar{\beta}_0 = K^{-1} \sum_{k=1}^K \beta_{0k}$ and $\bar{\beta}_i = K^{-1} \sum_{k=1}^K \beta_{i k}$. At the n data points, $\{\mathbf{x}_i, i = 1, 2, \dots, n\}$, (11.87) in matrix notation is given by

$$\left(\sum_{k=1}^K \beta_{0k} \right) \mathbf{1}_n + \mathbf{K} \left(\sum_{k=1}^K \beta_{\cdot k} \right) = \mathbf{0}, \quad (11.88)$$

where $\mathbf{K} = (K(\mathbf{x}_i, \mathbf{x}_j))$ is an $(n \times n)$ Gram matrix and $\beta_{\cdot k} = (\beta_{1k}, \dots, \beta_{nk})^\tau$. Let $\beta_{0k}^* = \beta_{0k} - \bar{\beta}_0$ and $\beta_{i k}^* = \beta_{i k} - \bar{\beta}_i$. Using (11.87), we see that the centered version of (11.84) is $f_k^*(\mathbf{x}_i) = \beta_{0k}^* + \sum_{\ell=1}^n \beta_{\ell k}^* K(\mathbf{x}_\ell, \mathbf{x}_i) = f_k(\mathbf{x}_i)$. Then,

$$\sum_{k=1}^K \|h_k^*(\cdot)\|^2 = \sum_{k=1}^K \beta_{\cdot k}^\tau \mathbf{K} \beta_{\cdot k} - K \bar{\beta}^\tau \mathbf{K} \bar{\beta} \leq \sum_{k=1}^K \beta_{\cdot k}^\tau \mathbf{K} \beta_{\cdot k} = \sum_{k=1}^K \|h_k(\cdot)\|^2, \quad (11.89)$$

where $\bar{\beta} = (\bar{\beta}_1, \dots, \bar{\beta}_n)^\tau$; if $\mathbf{K} \bar{\beta} = \mathbf{0}$, the inequality becomes an equality and so $\sum_{k=1}^K \beta_{0k} = 0$. Thus,

$$0 = K^2 \bar{\beta}^\tau \mathbf{K} \bar{\beta} = \left\| \sum_{i=1}^n \left(\sum_{k=1}^K \beta_{i k} \right) K(\mathbf{x}_i, \cdot) \right\|^2 = \left\| \sum_{k=1}^K \sum_{i=1}^n \beta_{i k} K(\mathbf{x}_i, \cdot) \right\|^2, \quad (11.90)$$

whence, $\sum_{k=1}^K \sum_{i=1}^n \beta_{i k} K(\mathbf{x}_i, \mathbf{x}) = 0$, for all $\mathbf{x}$. Thus,

$$\sum_{k=1}^K \left\{ \beta_{0k} + \sum_{i=1}^n \beta_{i k} K(\mathbf{x}_i, \mathbf{x}) \right\} = 0, \quad (11.91)$$

for every $\mathbf{x}$. So, minimizing (11.77) under the zero-sum constraint (11.76) only at the n data points is equivalent to minimizing (11.77) under the same constraint for every $\mathbf{x}$.

We next construct a Lagrangian formulation of the optimization problem (11.77) using the following notation. Let $\boldsymbol{\xi}_i = (\xi_{i1}, \dots, \xi_{iK})^\tau$ be a K -vector of slack variables corresponding to $(f(\mathbf{x}_i) - y_i)_+$, $i = 1, 2, \dots, n$, and let $(\boldsymbol{\xi}_{\cdot 1}, \dots, \boldsymbol{\xi}_{\cdot K}) = (\boldsymbol{\xi}_1, \dots, \boldsymbol{\xi}_n)^\tau$ be the $(n \times K)$ -matrix whose k th column is $\boldsymbol{\xi}_{\cdot k}$ and whose i th row is $\boldsymbol{\xi}_i$. Let $(\mathbf{L}_1, \dots, \mathbf{L}_K) = (\mathbf{L}(\mathbf{y}_1), \dots, \mathbf{L}(\mathbf{y}_n))^\tau$ be the $(n \times K)$ -matrix whose k th column is $\mathbf{L}_k$ and whose i th row is $\mathbf{L}(\mathbf{y}_i) = (L_{i1}, \dots, L_{iK})$. Let $(\mathbf{y}_{\cdot 1}, \dots, \mathbf{y}_{\cdot K}) = (\mathbf{y}_1, \dots, \mathbf{y}_n)^\tau$ denote the $(n \times K)$ -matrix whose k th column is $\mathbf{y}_{\cdot k}$ and whose i th row is $\mathbf{y}_i$.

The primal problem is to find $\{\beta_{0k}\}$, $\{\boldsymbol{\beta}_{\cdot k}\}$, and $\{\boldsymbol{\xi}_{\cdot k}\}$ to

$$\text{minimize} \quad \sum_{k=1}^K \mathbf{L}_k^\tau \boldsymbol{\xi}_{\cdot k} + \frac{n\lambda}{2} \sum_{k=1}^K \boldsymbol{\beta}_{\cdot k}^\tau \mathbf{K} \boldsymbol{\beta}_{\cdot k} \quad (11.92)$$

subject to

$$\beta_{0k} \mathbf{1}_n + \mathbf{K} \boldsymbol{\beta}_{\cdot k} - \mathbf{y}_{\cdot k} \leq \boldsymbol{\xi}_{\cdot k}, \quad k = 1, 2, \dots, K, \quad (11.93)$$

$$\boldsymbol{\xi}_{\cdot k} \geq \mathbf{0}, \quad k = 1, 2, \dots, K, \quad (11.94)$$

$$\left(\sum_{k=1}^K \beta_{0k} \right) \mathbf{1}_n + \mathbf{K} \left(\sum_{k=1}^K \boldsymbol{\beta}_{\cdot k} \right) = \mathbf{0}. \quad (11.95)$$

Form the primal functional $F_P = F_P(\{\beta_{0k}\}, \{\boldsymbol{\beta}_{\cdot k}\}, \{\boldsymbol{\xi}_{\cdot k}\})$, where

$$\begin{aligned} F_P &= \sum_{k=1}^K \mathbf{L}_k^\tau \boldsymbol{\xi}_{\cdot k} + \frac{n\lambda}{2} \sum_{k=1}^K \boldsymbol{\beta}_{\cdot k}^\tau \mathbf{K} \boldsymbol{\beta}_{\cdot k} \\ &\quad + \sum_{k=1}^K \boldsymbol{\alpha}_{\cdot k}^\tau (\beta_{0k} \mathbf{1}_n + \mathbf{K} \boldsymbol{\beta}_{\cdot k} - \mathbf{y}_{\cdot k} - \boldsymbol{\xi}_{\cdot k}) \\ &\quad - \sum_{k=1}^K \boldsymbol{\gamma}_k^\tau \boldsymbol{\xi}_{\cdot k} + \boldsymbol{\delta}^\tau \left(\left(\sum_{k=1}^K \beta_{0k} \right) \mathbf{1}_n + \mathbf{K} \left(\sum_{k=1}^K \boldsymbol{\beta}_{\cdot k} \right) \right). \end{aligned} \quad (11.96)$$

In (11.96), $\boldsymbol{\alpha}_{\cdot k} = (\alpha_{1k}, \dots, \alpha_{nk})^\tau$ and $\boldsymbol{\gamma}_k$ are n -vectors of nonnegative Lagrange multipliers for the inequality constraints (11.93) and (11.94), respectively, and $\boldsymbol{\delta}$ is an n -vector of unconstrained Lagrange multipliers for the equality constraint (11.95).

Differentiating (11.96) with respect to β_{0k} , $\boldsymbol{\beta}_{\cdot k}$, and $\boldsymbol{\xi}_{\cdot k}$ yields

$$\frac{\partial F_P}{\partial \beta_{0k}} = (\boldsymbol{\alpha}_{\cdot k} + \boldsymbol{\delta})^\tau \mathbf{1}_n, \quad (11.97)$$

$$\frac{\partial F_P}{\partial \boldsymbol{\beta}_{\cdot k}} = n\lambda \mathbf{K} \boldsymbol{\beta}_{\cdot k} + \mathbf{K} \boldsymbol{\alpha}_{\cdot k} + \mathbf{K} \boldsymbol{\delta}, \quad (11.98)$$

$$\frac{\partial F_P}{\partial \boldsymbol{\xi}_{\cdot k}} = \mathbf{L}_k - \boldsymbol{\alpha}_{\cdot k} - \boldsymbol{\gamma}_k, \quad (11.99)$$

$$\boldsymbol{\alpha}_{\cdot k} \geq \mathbf{0}, \quad (11.100)$$

$$\boldsymbol{\gamma}_k \geq \mathbf{0}. \quad (11.101)$$

The Karush–Kuhn–Tucker complementarity conditions are

$$\boldsymbol{\alpha}_{\cdot k}(\beta_{0k}\mathbf{1}_n + \mathbf{K}\boldsymbol{\beta}_{\cdot k} - \mathbf{y}_{\cdot k} - \boldsymbol{\xi}_{\cdot k})^\tau = 0, \quad k = 1, 2, \dots, K, \quad (11.102)$$

$$\boldsymbol{\gamma}_k \boldsymbol{\xi}_{\cdot k}^\tau = 0, \quad k = 1, 2, \dots, K, \quad (11.103)$$

where, from (11.99), $\boldsymbol{\gamma}_k = \mathbf{L}_k - \boldsymbol{\alpha}_{\cdot k}$. Note that (11.102) and (11.103) are outer products of two column vectors, meaning that each of the n^2 elementwise products of those vectors are zero.

From (11.99) and (11.101), we have that $\mathbf{0} \leq \boldsymbol{\alpha}_{\cdot k} \leq \mathbf{L}_k$, $k = 1, 2, \dots, K$. Suppose, for some i , $0 < \alpha_{ik} < L_{ik}$; then, $\gamma_{ik} > 0$, and, from (11.103), $\xi_{ik} = 0$, whence, from (11.102), $y_{ik} = \beta_{0k} + \sum_{\ell=1}^n \beta_{\ell k} K(\mathbf{x}_\ell, \mathbf{x}_i)$.

Setting the derivatives equal to zero for $k = 1, 2, \dots, K$ yields $\boldsymbol{\delta} = -\bar{\boldsymbol{\alpha}} = -K^{-1} \sum_{k=1}^K \boldsymbol{\alpha}_{\cdot k}$ from (11.97), whence, $(\boldsymbol{\alpha}_{\cdot k} - \bar{\boldsymbol{\alpha}})^\tau \mathbf{1}_n = 0$, and, from (11.98), $\boldsymbol{\beta}_{\cdot k} = -(n\lambda)^{-1}(\boldsymbol{\alpha}_{\cdot k} - \bar{\boldsymbol{\alpha}})$, assuming that $\mathbf{K}$ is positive-definite. If $\mathbf{K}$ is not positive-definite, then $\boldsymbol{\beta}_{\cdot k}$ is not uniquely determined. Because (11.97), (11.98), and (11.99) are each zero, we construct the dual functional F_D by using them to remove a number of the terms of F_P .

The resulting dual problem is to find $\{\boldsymbol{\alpha}_{\cdot k}\}$ to

$$\text{minimize } F_D = \frac{1}{2} \sum_{k=1}^K (\boldsymbol{\alpha}_{\cdot k} - \bar{\boldsymbol{\alpha}})^\tau \mathbf{K} (\boldsymbol{\alpha}_{\cdot k} - \bar{\boldsymbol{\alpha}}) + n\lambda \sum_{k=1}^K \boldsymbol{\alpha}_{\cdot k}^\tau \mathbf{y}_{\cdot k} \quad (11.104)$$

$$\text{subject to } \mathbf{0} \leq \boldsymbol{\alpha}_{\cdot k} \leq \mathbf{L}_k, \quad k = 1, 2, \dots, K, \quad (11.105)$$

$$(\boldsymbol{\alpha}_{\cdot k} - \bar{\boldsymbol{\alpha}})^\tau \mathbf{1}_n = 0, \quad k = 1, 2, \dots, K. \quad (11.106)$$

From the solution, $\{\hat{\boldsymbol{\alpha}}_{\cdot k}\}$, to this quadratic programming problem, we set

$$\hat{\boldsymbol{\beta}}_{\cdot k} = -(n\lambda)^{-1}(\hat{\boldsymbol{\alpha}}_{\cdot k} - \hat{\boldsymbol{\alpha}}), \quad (11.107)$$

where $\hat{\boldsymbol{\alpha}} = K^{-1} \sum_{k=1}^K \hat{\boldsymbol{\alpha}}_{\cdot k}$.

The multiclass classification solution for a new $\mathbf{x}$ is given by

$$C_k(\mathbf{x}) = \arg \max_k \{\hat{f}_k(\mathbf{x})\}, \quad (11.108)$$

where

$$\hat{f}_k(\mathbf{x}) = \hat{\beta}_{0k} + \sum_{\ell=1}^n \hat{\beta}_{\ell k} K(\mathbf{x}_\ell, \mathbf{x}), \quad k = 1, 2, \dots, K. \quad (11.109)$$

Suppose the row vector $\hat{\alpha}_i = (\hat{\alpha}_{i1}, \dots, \hat{\alpha}_{iK}) = \mathbf{0}$ for $(\mathbf{x}_i, \mathbf{y}_i)$; then, from (11.107), $\hat{\beta}_i = (\hat{\beta}_{i1}, \dots, \hat{\beta}_{iK}) = \mathbf{0}$. It follows that the term $\hat{\beta}_{ik}K(\mathbf{x}_i, \mathbf{x}) = 0$, $k = 1, 2, \dots, K$. Thus, any term involving $(\mathbf{x}_i, \mathbf{y}_i)$ does not appear in (11.109); in other words, it does not matter whether $(\mathbf{x}_i, \mathbf{y}_i)$ is or is not included in the learning set $\mathcal{L}$ because it has no effect on the solution. This result leads us to a definition of support vectors: an observation $(\mathbf{x}_i, \mathbf{y}_i)$ is called a *support vector* if $\hat{\beta}_i = (\hat{\beta}_{i1}, \dots, \hat{\beta}_{iK}) \neq \mathbf{0}$. As in the binary SVM solution, it is in our computational best interests for there to be relatively few support vectors for any given application.

The one issue remaining is the choice of tuning parameter λ (and any other parameters involved in the computation of the kernel). A *generalized approximate cross-validation* (GACV) method is derived in Lee, Lin, and Wahba (2004) based upon an approximation to the leave-one-out cross-validation technique used for penalized-likelihood methods. The basic idea behind GACV is the following. Write (11.77) as

$$I_\lambda(\mathbf{f}, \mathcal{Y}) = n^{-1} \sum_{i=1}^n g(\mathbf{y}_i, \mathbf{f}(\mathbf{x}_i)) + J_\lambda(\mathbf{f}), \quad (11.110)$$

where $g(\mathbf{y}_i, \mathbf{f}(\mathbf{x}_i)) = [\mathbf{L}(\mathbf{y}_i)]^\tau (\mathbf{f}(\mathbf{x}_i) - \mathbf{y}_i)_+$ and $J_\lambda(\mathbf{f}) = (\lambda/2) \sum_{i=1}^n \|h_j\|^2$. Let $\mathbf{f}_\lambda = \arg \min_{\mathbf{f}} I_\lambda(\mathbf{f}, \mathcal{Y})$ and let $\mathbf{f}_\lambda^{(-i)}$ denote that $\mathbf{f}_\lambda$ that yields the minimum of $I_\lambda(\mathbf{f}, \mathcal{Y})$ by omitting the i th observation $(\mathbf{x}_i, \mathbf{y}_i)$ from the first term in (11.110). If we write

$$g(\mathbf{y}_i, \mathbf{f}_\lambda^{(-i)}(\mathbf{x}_i)) = g(\mathbf{y}_i, \mathbf{f}_\lambda(\mathbf{x}_i)) + [g(\mathbf{y}_i, \mathbf{f}_\lambda^{(-i)}(\mathbf{x}_i)) - g(\mathbf{y}_i, \mathbf{f}_\lambda(\mathbf{x}_i))], \quad (11.111)$$

then the λ that minimizes $n^{-1} \sum_{i=1}^n g(\mathbf{y}_i, \mathbf{f}_\lambda^{(-i)}(\mathbf{x}_i))$ is found by using a suitable approximation of $D(\lambda) = n^{-1} \sum_{i=1}^n [g(\mathbf{y}_i, \mathbf{f}_\lambda^{(-i)}(\mathbf{x}_i)) - g(\mathbf{y}_i, \mathbf{f}_\lambda(\mathbf{x}_i))]$, computed over a grid of values of λ .

This solution of the multiclass SVM problem has been found to be successful in simulations and in analyzing real data. Comparisons of various multiclass classification methods, such as multiclass SVM, “all-versus-rest,” LDA, and QDA, over a number of data sets show that no one classification method appears to be superior for all situations studied; performance appears to depend upon the idiosyncracies of the data to be analyzed.

11.5 Support Vector Regression

The SVM was designed for classification. Can we extend (or generalize) the idea to regression? How would the main concepts used in SVM — convex optimization, optimal separating hyperplane, support vectors, margin, sparseness of the solution, slack variables, and the use of kernels — translate to the regression situation? It turns out that all of these concepts find

their analogues in regression analysis and they add a different view to the topic than the views we saw in Chapter 5.

11.5.1 ϵ -Insensitive Loss Functions

In SVM classification, the margin is used to determine the amount of separation between two nonoverlapping classes of points: the bigger the margin, the more confident we are that the optimal separating hyperplane is a superior classifier. In regression, we are not interested in separating points but in providing a function of the input vectors that would track the points closely. Thus, a regression analogue for the margin would entail forming a “band” or “tube” around the true regression function that contains most of the points. Points not contained within the tube would be described through slack variables. In formulating these ideas, we first need to define an appropriate loss function.

We define a loss function that ignores errors associated with points falling within a certain distance (e.g., $\epsilon > 0$) of the true linear regression function,

$$\mu(\mathbf{x}) = \beta_0 + \mathbf{x}^\tau \boldsymbol{\beta}. \quad (11.112)$$

In other words, if the point $(\mathbf{x}, y)$ is such that $|y - \mu(\mathbf{x})| \leq \epsilon$, then the loss is taken to be zero; if, on the other hand, $|y - \mu(\mathbf{x})| > \epsilon$, then we take the loss to be $|y - \mu(\mathbf{x})| - \epsilon$.

With this strategy in mind, we can define the following two types of loss function:

- $L_1^\epsilon(y, \mu(\mathbf{x})) = \max\{0, |y - \mu(\mathbf{x})| - \epsilon\},$
- $L_2^\epsilon(y, \mu(\mathbf{x})) = \max\{0, (y - \mu(\mathbf{x}))^2 - \epsilon\}.$

The first loss function, L_1^ϵ , is called the *linear ϵ -insensitive loss function*, and the second, L_2^ϵ , is the *quadratic ϵ -insensitive loss function*. The two loss functions, linear (red curve) and quadratic (blue curve), are graphed in Figure 11.5. We see that the linear loss function ignores all errors falling within $\pm\epsilon$ of the true regression function $\mu(\mathbf{x})$ while dampening in a linear fashion errors that fall outside those limits.

11.5.2 Optimization for Linear ϵ -Insensitive Loss

We define slack variables ξ_i and ξ'_j in the following way. If the point $(\mathbf{x}_i, y_i)$ lies above the ϵ -tube, then $\xi'_i = y_i - \mu(\mathbf{x}_i) - \epsilon \geq 0$, whereas if the point $(\mathbf{x}_j, y_j)$ lies below the ϵ -tube, then $\xi_j = \mu(\mathbf{x}_j) - \epsilon - y_j \geq 0$. For points that fall outside the ϵ -tube, the values of the slack variables depend

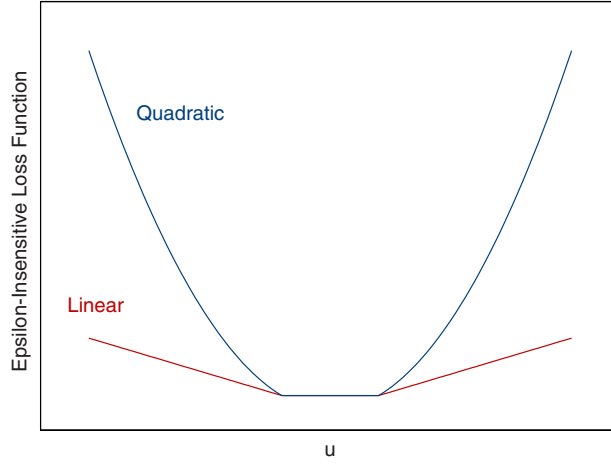


FIGURE 11.5. The linear ϵ -insensitive loss function (red curve) and the quadratic ϵ -insensitive loss function (blue curve) for support vector regression. Plotted are $L_i(u) = \max\{0, |u|^i - \epsilon\}$ vs. u , $i = 1, 2$, where $u = y - \mu(\mathbf{x})$. For the linear loss function, the “flat” part of the curve has width 2ϵ .

upon the shape of the loss function; for points inside the ϵ -tube, the slack variables have value zero.

For linear ϵ -insensitive loss, the primal optimization problem is to find β_0 , β , $\xi' = (\xi'_1, \dots, \xi'_n)^\tau$, and $\xi = (\xi_1, \dots, \xi_n)^\tau$ to

$$\text{minimize} \quad \frac{1}{2} \|\beta\|^2 + C \sum_{i=1}^n (\xi_i + \xi'_i) \quad (11.113)$$

$$\begin{aligned} \text{subject to} \quad & y_i - (\beta_0 + \mathbf{x}_i^\tau \beta) \leq \epsilon + \xi'_i, \\ & (\beta_0 + \mathbf{x}_i^\tau \beta) - y_i \leq \epsilon + \xi_i, \\ & \xi'_i \geq 0, \xi_i \geq 0, \quad i = 1, 2, \dots, n. \end{aligned} \quad (11.114)$$

The constant $C > 0$ exists to balance the flatness of the function μ against our tolerance of deviations larger than ϵ . Notice that because ϵ is found only in the constraints, the solution to this optimization problem has to incorporate a band around the regression function.

Form the primal Lagrangian,

$$\begin{aligned} F_P = \quad & \frac{1}{2} \|\beta\|^2 + C \sum_{i=1}^n (\xi_i + \xi'_i) - \sum_i a_i \{y_i - (\beta_0 + \mathbf{x}_i^\tau \beta) - \epsilon - \xi'_i\} \\ & - \sum_i b_i \{(\beta_0 + \mathbf{x}_i^\tau \beta) - y_i - \epsilon - \xi_i\} \\ & - \sum_i c_i \xi'_i - \sum_i d_i \xi_i, \end{aligned} \quad (11.115)$$

where a_i, b_i, c_i , and d_i , $i = 1, 2, \dots, n$, are the Lagrange multipliers. This, in turn, implies that a_i, b_i, c_i, d_i , $i = 1, 2, \dots, n$, are all nonnegative. The derivatives are

$$\frac{\partial F_P}{\partial \beta_0} = \sum_i a_i - \sum_i b_i \quad (11.116)$$

$$\frac{\partial F_P}{\partial \boldsymbol{\beta}} = \boldsymbol{\beta} + \sum_i a_i \mathbf{x}_i - \sum_i b_i \mathbf{x}_i \quad (11.117)$$

$$\frac{\partial F_P}{\partial \xi_i} = C + b_i - d_i \quad (11.118)$$

$$\frac{\partial F_P}{\partial \xi'_i} = C + a_i - c_i \quad (11.119)$$

Setting these derivatives equal to zero for a stationary solution yields:

$$\boldsymbol{\beta}^* = \sum_i (b_i - a_i) \mathbf{x}_i, \quad (11.120)$$

$$\sum_i (b_i - a_i) = 0, \quad (11.121)$$

$$C + b_i - d_i = 0, \quad C + a_i - c_i = 0, \quad i = 1, 2, \dots, n. \quad (11.122)$$

The expression (11.120) is known as the *support vector expansion* because $\boldsymbol{\beta}^*$ can be written as a linear combination of the input vectors $\{\mathbf{x}_i\}$. Setting $\boldsymbol{\beta} = \boldsymbol{\beta}^*$ in the true regression equation (11.112) gives us

$$\mu^*(\mathbf{x}) = \beta_0 + \sum_{i=1}^n (b_i - a_i) (\mathbf{x}^\tau \mathbf{x}_i). \quad (11.123)$$

Substituting $\boldsymbol{\beta}^*$ into the primal Lagrangian and using (11.120) and (11.121) gives us the dual problem: find $\mathbf{a} = (a_1, \dots, a_n)^\tau$, $\mathbf{b} = (b_1, \dots, b_n)^\tau$ to

$$\begin{aligned} \text{maximize} \quad F_D &= (\mathbf{b} - \mathbf{a})^\tau \mathbf{y} - \epsilon (\mathbf{b} + \mathbf{a})^\tau \mathbf{1}_n \\ &\quad - \frac{1}{2} (\mathbf{b} - \mathbf{a})^\tau \mathbf{K} (\mathbf{b} - \mathbf{a}) \end{aligned} \quad (11.124)$$

$$\text{subject to} \quad \mathbf{0} \leq \mathbf{a}, \mathbf{b} \leq C \mathbf{1}_n, \quad (\mathbf{b} - \mathbf{a})^\tau \mathbf{1}_n = 0, \quad (11.125)$$

where $\mathbf{K} = (\langle \mathbf{x}_i, \mathbf{x}_j \rangle)$ for linear SVM. The Karush–Kuhn–Tucker complementarity conditions state that the products of the dual variables and the constraints are all zero:

$$a_i (\beta_0 + \mathbf{x}_i^\tau \boldsymbol{\beta} - y_i - \epsilon - \xi_i) = 0, \quad i = 1, 2, \dots, n, \quad (11.126)$$

$$b_i (y_i - \beta_0 - \mathbf{x}_i^\tau \boldsymbol{\beta} - \epsilon - \xi'_i) = 0, \quad i = 1, 2, \dots, n, \quad (11.127)$$

$$\xi_i \xi'_i = 0, \quad a_i b_i = 0, \quad i = 1, 2, \dots, n, \quad (11.128)$$

$$(a_i - C) \xi_i = 0, \quad (b_i - C) \xi'_i = 0, \quad i = 1, 2, \dots, n. \quad (11.129)$$